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Set 16: Customer Demographics Analysis

Aim

To analyse customer demographics in R by creating a bar chart of customer ages, a pie chart of gender distribution, a demographic table, and a combined dashboard.

Questions (combined into one experiment)

1. Bar chart of customer ages (axes labelled, titled)
2. Pie chart of customers by gender
3. Table of demographic information for each customer
4. Dashboard combining the bar chart, pie chart and table

Procedure

1. Open RStudio and install/load the packages ggplot2 and gridExtra.
2. Create the dataset as a data frame.
3. Draw the charts for Questions 1 and 2 using ggplot2 with titles and axis labels.
4. Display the dataset as a table using print() and tableGrob().
5. Combine the two charts and the table into a dashboard using grid.arrange().
6. Run the program and observe the output.

Code

```
library(ggplot2)
library(gridExtra)

# Dataset
df <- data.frame(CustomerID = 1:3, Age = c(28, 35, 42),
                 Gender = c("Female", "Male", "Female"),
                 Income = c(50000, 60000, 75000))

# Q1. Bar chart of customer ages
p1 <- ggplot(df, aes(x = factor(Age))) + geom_bar(fill = "steelblue") +
  labs(title = "Distribution of Customer Ages", x = "Age", y = "Number of Customers")
print(p1)

# Q2. Pie chart of gender
g <- as.data.frame(table(df$Gender)); names(g) <- c("Gender", "Count")
p2 <- ggplot(g, aes(x = "", y = Count, fill = Gender)) + geom_col() +
  coord_polar("y") + theme_void() + labs(title = "Customers by Gender")
print(p2)

# Q3. Table of demographics
```

```
print(df)
tb <- tableGrob(df, rows = NULL)

# Q4. Dashboard (bar chart + pie chart + table)
grid.arrange(p1, p2, tb, ncol = 2, top = "Customer Demographics Dashboard")
```

Output

Table (Question 3):

	CustomerID	Age	Gender	Income
1	1	28	Female	50000
2	2	35	Male	60000
3	3	42	Female	75000

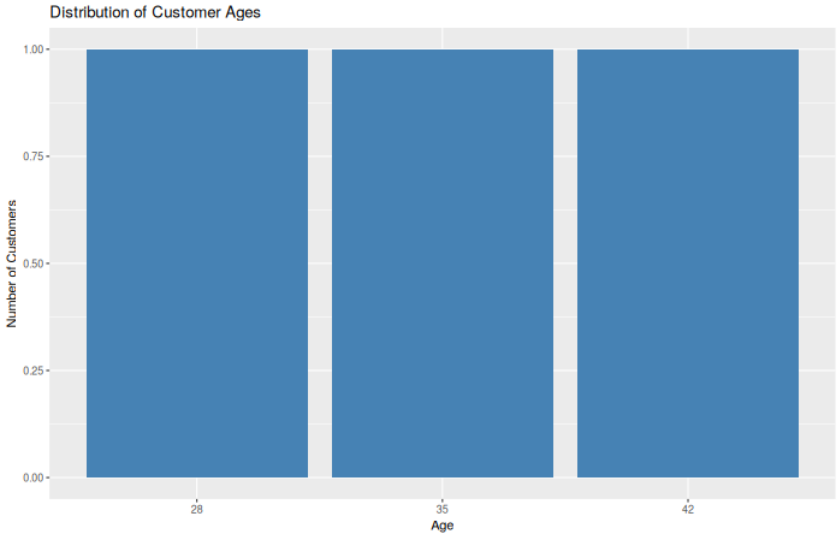


Figure 1: Question 1

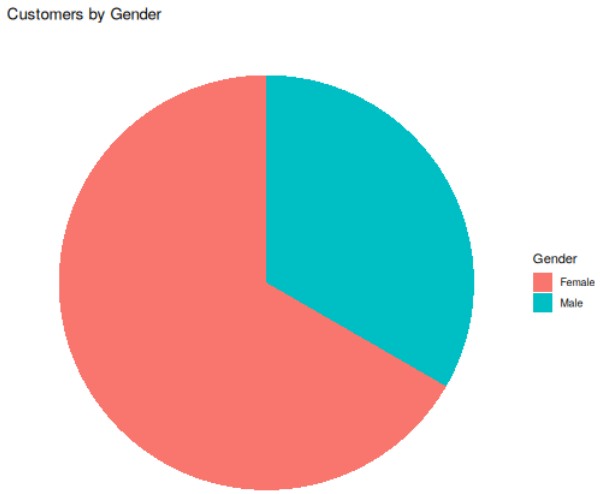
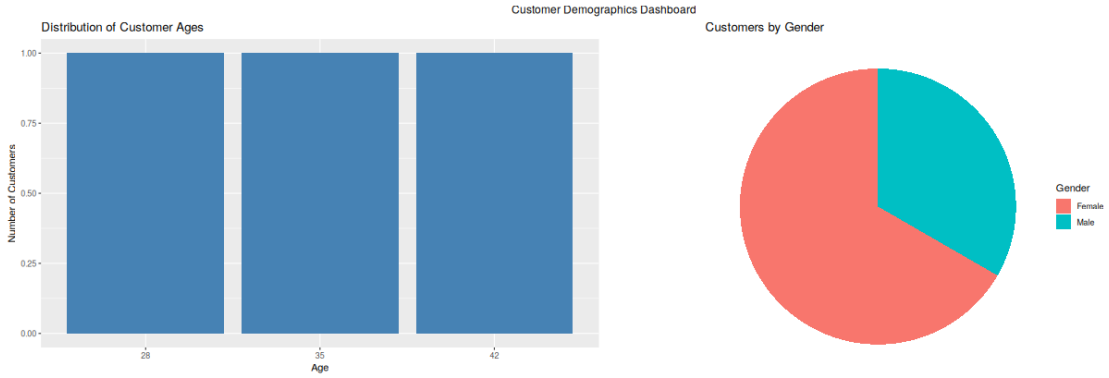


Figure 2: Question 2



CustomerID	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Figure 3: Dashboard (Question 4)

Result

The bar chart, pie chart, demographic table and combined dashboard for customer demographics were created successfully using R.